VANISHING OF LOCAL COHOMOLOGY OUTSIDE THE KOSZUL DEGREE FOR FLAT QUASI-COHERENT MODULES

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Locally the sections with support R\mathcal{H}_Z(F) are given by the extended alternating Cech complex of the Koszul regular sequence. invoked at proof lines 1247-1248. Reduces the statement to an algebra statement about the Cech complex.
- For a Koszul regular sequence in a ring, the extended alternating Cech complex of the ring itself only has cohomology in degree c. invoked at proof lines 1256-1259. Gives the vanishing over R.
- The extended alternating Cech complex of a module M is obtained by tensoring the complex of R with M. invoked at proof lines 1259-1262. With M flat this preserves cohomology, concluding the vanishing for M and hence for F.

Source reference. Original lemma 0G7M: sources/stacks-perfect/pinned-original-source.tex, source lines 1185–1185. Cited at proof lines 1247-1248 for the reduction to the algebra statement; the vanishing argument is inside the counted span.

Source reference. Original section 0G6F: sources/stacks-chow/tags.txt, source lines 20195–20195. Cited at proof lines 1255-1255 as the source of the complex under study; the vanishing argument is inside the counted span.

Source reference. Original lemma 0G6L: sources/stacks-chow/tags.txt, source lines 20201–20201. Cited at proof lines 1256-1258 for the vanishing over R; the transfer to M is inside the counted span.

Source reference. Original lemma 0G6H: sources/stacks-chow/tags.txt, source lines 20197–20197. Cited at proof lines 1261-1262 to identify the complex for M; the flatness conclusion is inside the counted span.

Lemma 0.2. *Let X be a scheme. Let $Z \subset X$ be a closed subset which can locally be cut out by a Koszul regular sequence having c elements. Then $\mathcal{H}^i_Z(\mathcal{F}) = 0$ for $i \neq c$ for every flat, quasi-coherent $\mathcal{O}_X$-module $\mathcal{F}$.*

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Proof. By the description of $R\mathcal{H}_Z(\mathcal{F})$ given in Lemma 0G7M this boils down to the following algebra statement: given a ring R, a Koszul regular sequence $f_1, \ldots, f_c \in R$, and a flat R-module M, the extended alternating Čech complex $M \to \bigoplus_{i_0} M_{f_{i_0}} \to \bigoplus_{i_0<i_1} M_{f_{i_0}f_{i_1}} \to \ldots \to M_{f_1 \ldots f_c}$ from More on Algebra, Section 0G6F only has cohomology in degree c. By More on Algebra, Lemma 0G6L we obtain the desired vanishing for the extended alternating Čech complex of R. Since the complex for M is obtained by tensoring this with the flat R-module M (More on Algebra, Lemma 0G6H) we conclude. □